R-Square

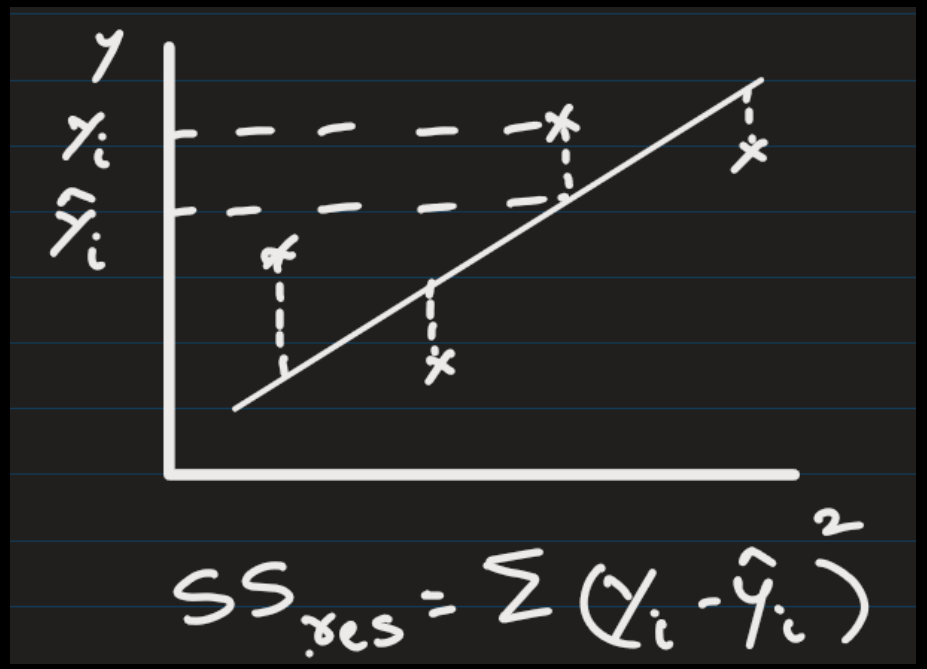
Coefficient of Determination (R^2 Score):

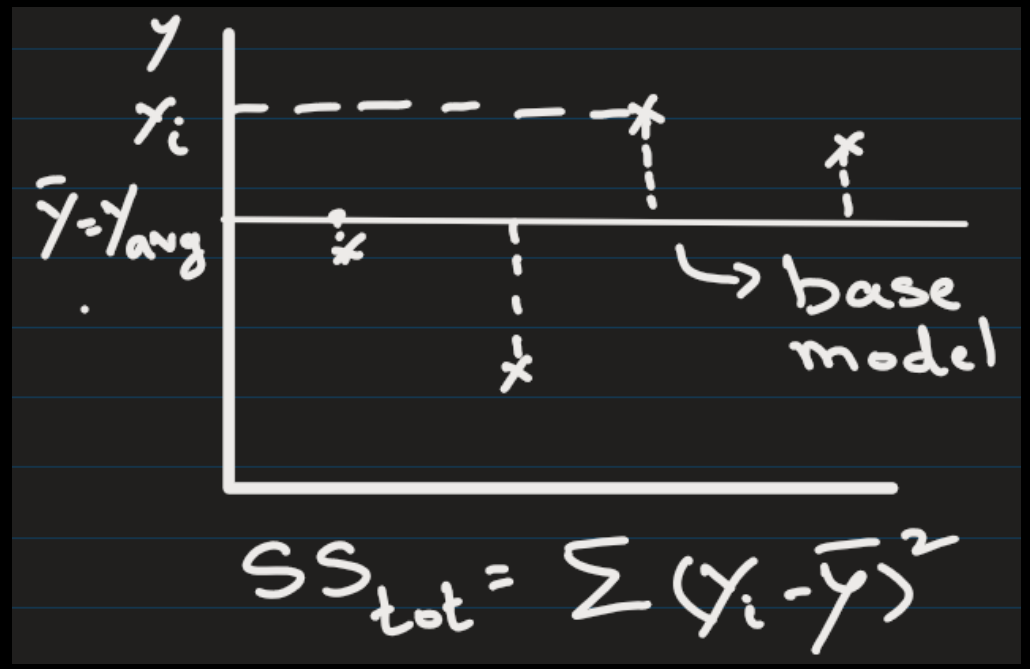
R-square measures how much of the **variance in Y is explained by the model.**

$$R^2 = 1 - \frac{SS_{res}}{SS_{tot}}$$

$SS_{res} = \sum(Y_i - \hat{Y}_i)^2 \rightarrow$ Residual Sum of Squares

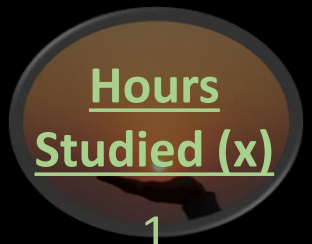
$SS_{tot} = \sum(Y_i - \bar{Y})^2 \rightarrow$ Total Sum of Squares





Hours Studied (x)	Score (y)	Predicted ($\hat{Y} = -5 + 12x$)
1	5	**7**
2	30	**19**
3	20	**31**
4	40	**43**
5	60	**55**

What is the R-square value ?



Hours Studied (x)	Score (y)	Predicted ($\hat{Y} = -5 + 12x$)	$e_i = Y_i - \hat{Y}_i$	$Y_i - \bar{Y}$
1	5	7	e_1 = 5 – 7 = - 2	5 – 31 = -26
2	30	19	e_2 = 30 – 19 = + 11	30 – 31 = -1
3	20	31	e_3 = 20 – 31 = - 11	20 – 31 = -11
4	40	43	e_4 = 40 – 43 = - 3	40 – 31 = 9
5	60	55	e_5 = 60 – 55 = + 5	60 – 31 = 29
	$\bar{Y}$ = 31		RSS = 280	TSS = 1720

- Residual sum of squares (SSR or SS_{res}):

$$SS_{res} = \sum e_i^2 = 4 + 121 + 121 + 9 + 25 = 280$$

- Total sum of squares (SS_{tot}):

$$SS_{tot} = \sum (Y_i - \bar{Y})^2 = 676 + 1 + 121 + 81 + 841 = 1720$$

$$R^2 = 1 - \frac{SS_{res}}{SS_{tot}} = 1 - \frac{280}{1720} = 0.8372093$$

The model explains about **83.72%** of the variance in scores.

$$R^2 = 1 - \frac{SS_{res}}{SS_{tot}}$$

Ranges from -∞ to 1 (Higher is better)

R^2 = 1 → Perfect model (explains all variance).

R^2 = 0 → Model performs as badly as the mean prediction model (i.e. Mean of target).

R^2 < 0 → Model performs worse than a simple mean prediction (bad model).

Above 0.9 → Excellent

0.7 - 0.9 → Good

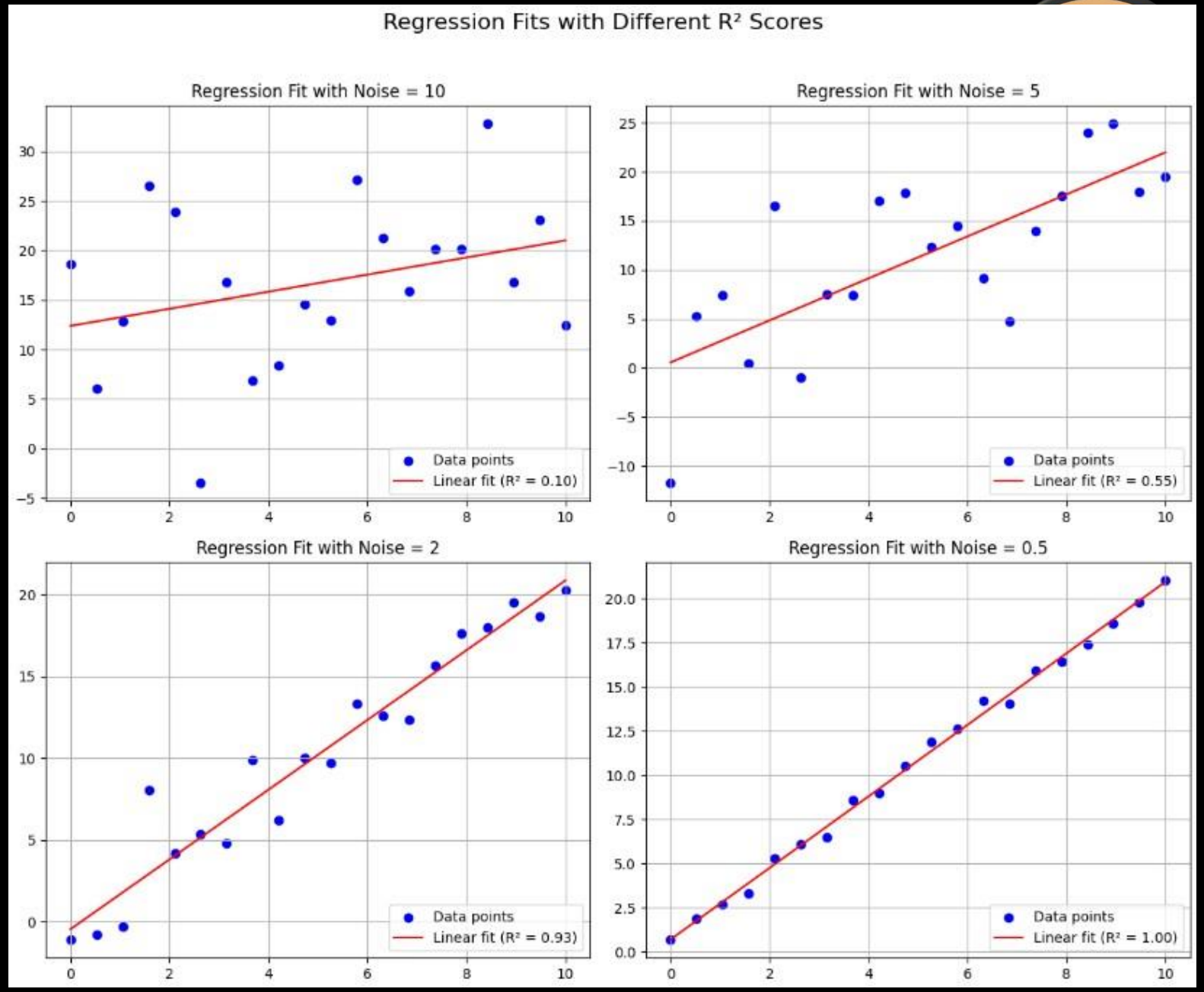
0.5 - 0.7 → Moderate

< 0.5 → Poor

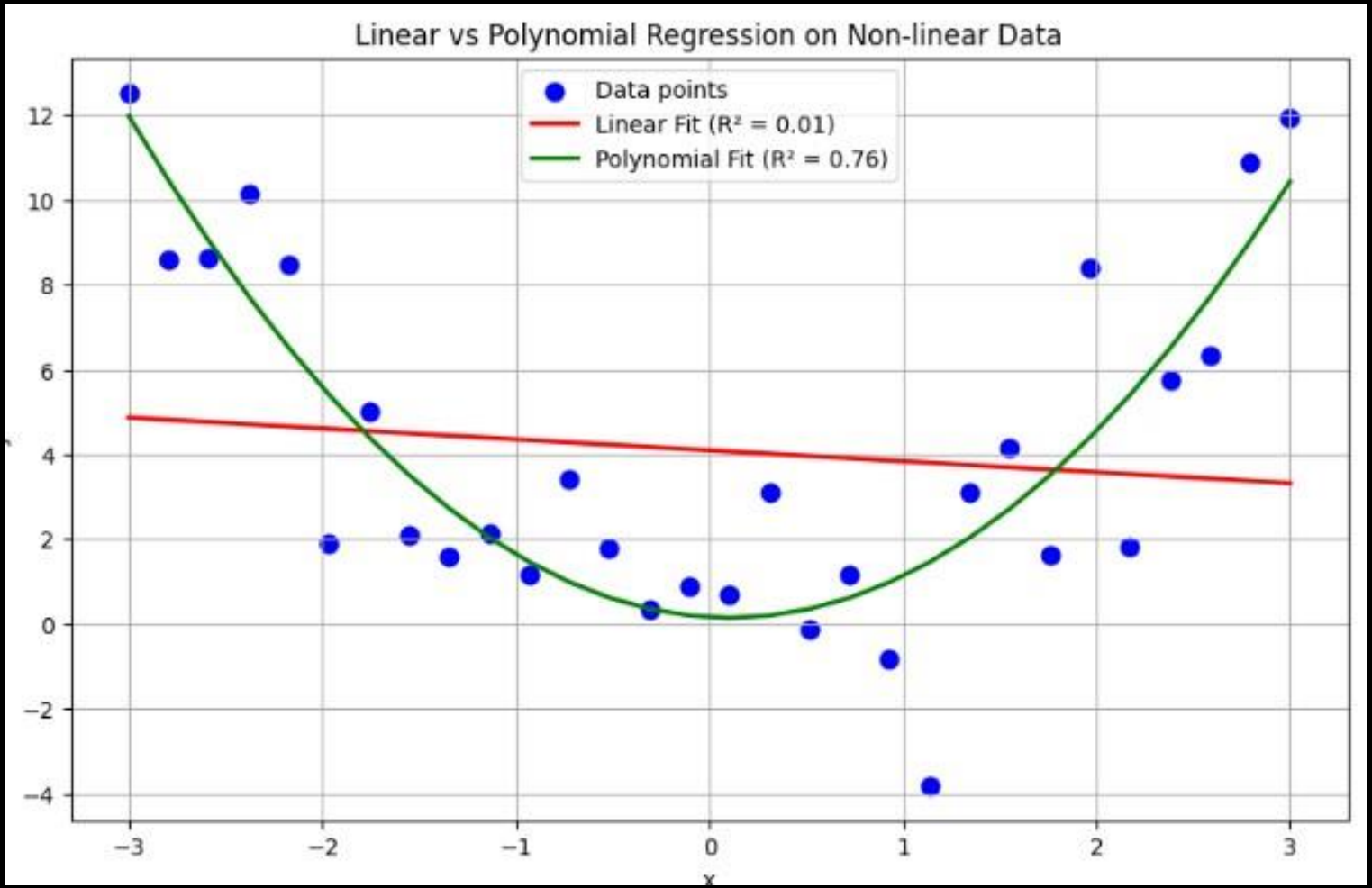
Hours Studied (x)	Score (y)	Baseline Predicted (Mean)
1	5	**31**
2	30	**31**
3	20	**31**
4	40	**31**
5	60	**31**

Mean = 31

Code from jupyter n/b



Code from jupyter n/b



When to use R-square ?

Use R-square when target scale is large

If target values are large (like revenue in millions, billions, etc), MAE might look large numerically. R^2 is scale-free, so easier to compare.

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